SOLUTION

SECTION–I

1. Ans. (D)

$$\tan(180^\circ+20^\circ)\left(\frac{\cos^2 10^\circ-\sin^2 10^\circ}{\cos 10^\circ \sin 10^\circ}\right)+1$$

$$= \tan 20^\circ\left(\frac{\cos 20^\circ}{\sin 20^\circ}.2\right)+1 = 2+1=3.$$

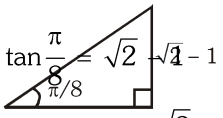
2. Ans. (D)

$$\frac{(\cos 1^\circ+\cos 89^\circ)(\cos 2^\circ+\cos 88^\circ)(\cos 3^\circ+\cos 87^\circ)\ldots(\cos 45^\circ+\cos 45^\circ)}{\cos 1^\circ.\cos 2^\circ\cos 3^\circ\ldots\ldots\cos 45^\circ}$$

$$=\frac{(2.\cos 45^\circ\cos 44^\circ)(2\cos 45^\circ\cos 43^\circ)(2\cos 45^\circ\cos 42^\circ)\ldots\ldots 2\cos 45^\circ}{\cos 1^\circ.\cos 2^\circ\cos 3^\circ\ldots\ldots\cos 45^\circ}$$

$$=(\sqrt{2})^{44}.2=2^{23} \quad \Rightarrow \quad a+b=25$$

3. Ans. (C)



$$\Rightarrow \sin\frac{\pi}{8}=\frac{\sqrt{2}-1}{\sqrt{4-2\sqrt{2}}}$$

$$=\frac{\sqrt{2}-1}{\sqrt{2\sqrt{2}(\sqrt{2}-1)}}$$

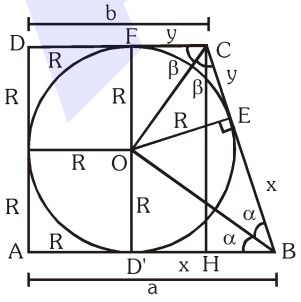
$$=\sqrt{\frac{\sqrt{2}-1}{2\sqrt{2}}}=\sqrt{\frac{1}{2}-\frac{1}{2\sqrt{2}}}=\sqrt{\frac{1}{\sqrt{4}}-\frac{1}{\sqrt{8}}}$$

$\Rightarrow m+n=12$

4. Ans. B

Let ABCD is a right angled trapezium.

Let radius of circle is R



& let BD = BE = x and
CE = CF = y

$\because$ AB = a

$R + x = a \Rightarrow x = a - R$ ..... (1)

& DC = b

$R + y = b \Rightarrow y = b - R$ ..... (2)

In Δ OD'B, $\tan\alpha = \frac{R}{x}$

Now in Δ CHB, $\sin 2\alpha = \frac{CH}{BC}$

$$\Rightarrow \frac{2\tan\alpha}{1+\tan^2\alpha}=\frac{2R}{x+y}$$

$$\Rightarrow \frac{\frac{2R}{x}}{1+\frac{R^2}{x^2}}=\frac{2R}{x+y}$$

$$\Rightarrow \frac{2Rx}{x^2+R^2}=\frac{2R}{x+y}$$

$\Rightarrow x^2+xy=x^2+R^2 \Rightarrow xy=R^2 \Rightarrow$

$(a-R)(b-R)=R^2$ [From (1) and (2)]

$\Rightarrow ab-(a+b)R+R^2=R^2$

$\Rightarrow ab-(a+b)R=0$

$$\therefore \quad R=\frac{ab}{a+b}$$

5. Ans. (D)

$\log_{1/4}\beta = -1 \Rightarrow \beta = 4$

$\log_\beta\alpha = -1 \Rightarrow \alpha = \frac{1}{\beta}=\frac{1}{4}$

$$\gamma=\log_\alpha 8=\log_{1/4}8=-\frac{3}{2}$$

Now $\frac{1}{\alpha}+1=4+1=5$

$$\beta^2+4\gamma^2=16+4\left(-\frac{3}{2}\right)^2=25$$

$$\therefore \quad \left(\frac{1}{\alpha}+1\right)^{\log_{\sqrt{5}}(\beta^2+4\gamma^2)}=5^{\log_{\sqrt{5}}25}=625$$

6. Ans. (A)

For any positive value of x LHS is positive and RHS is negative

$\therefore$ no value of x will satisfy the given equation

Hence there is no solution

7. Ans. (B)

$\sin\theta+\cos\theta=\sqrt{2}\sin(\theta+45^\circ)$

$=\sqrt{2}\sin(145^\circ)=\sqrt{2}\sin 35^\circ$ i.e. $\in (0,1)$

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8. Ans. (B)

$$\frac{\sin^2\theta.\sin\theta}{\cos\theta}+\frac{\cos^3\theta}{\sin\theta}=(1+\sin2\theta)+\frac{(\sin^2\theta+\cos^2\theta)}{\sin\theta.\cos\theta}$$

$$\Rightarrow \quad \frac{\sin^4\theta+\cos^4\theta}{\sin\theta.\cos\theta}=(1+\sin2\theta)+\frac{1}{\sin\theta.\cos\theta}$$

$$\Rightarrow \quad \frac{1-2\sin^2\theta\cos^2\theta}{\sin\theta.\cos\theta}=(1+\sin2\theta)+\frac{2}{(\sin2\theta)}$$

$$\Rightarrow \quad \frac{2\left(1-\frac{\sin^2 2\theta}{2}\right)}{\sin2\theta}=(1+\sin2\theta)+\frac{2}{\sin2\theta}$$

$$\Rightarrow \quad \frac{2\left(1-\frac{t^2}{2}\right)}{t}=(1+t)+\frac{2}{t} \quad \Rightarrow$$

$$\frac{(2-t^2)}{t}=\frac{t^2+t+2}{t}$$

$\Rightarrow \quad 2t^2+t=0 \Rightarrow t(2t+1)=0 \quad \Rightarrow \quad t=0$

& $\quad t=-\frac{1}{2}$

$\sin2\theta=0 \Rightarrow \theta=0$

or $\quad \sin2\theta=-\frac{1}{2}$

$$\Rightarrow 2\theta=\frac{7\pi}{6}\text{ or }\frac{11\pi}{6}$$

$$\Rightarrow \theta=\frac{7\pi}{12}\text{ or }\frac{11\pi}{12}$$

$\therefore$ sum of solution

$$=0+\frac{7\pi}{12}+\frac{11\pi}{12}$$

$$=\frac{3\pi}{2}$$

9. Ans. (C)

$$2^{\sec^2 x}\sqrt{\left(\tan y-\frac{1}{2}\right)^2+\frac{1}{4}}\le 1$$

$\Rightarrow \quad \sec^2 x\ge 1$ & because $\left(\tan y-\frac{1}{2}\right)^2+\frac{1}{4}\ge\frac{1}{4}$

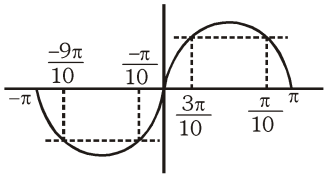
$\therefore \quad \sec^2 x=1$

$\Rightarrow \quad x=0,\ \pi,\ 2\pi$

10. Ans. (C)

$$-\sqrt{5}<4\sin x-1<\sqrt{5}$$

$$\Rightarrow \quad \frac{1-\sqrt{5}}{4}<\sin x<\frac{\sqrt{5}+1}{4}$$



$$\therefore \quad -\pi<x<-\frac{9\pi}{10},\ \frac{-\pi}{10}<x<\frac{3\pi}{10},\ \frac{7\pi}{10}<x<\pi$$

11. Ans. (A,B,C)

$$\sqrt{\cos2x}+\sqrt{1+\sin2x}=2\sqrt{\sin x+\cos x}$$

$$\Rightarrow \quad \sqrt{(\cos^2 x-\sin^2 x)}+\sqrt{(\cos x+\sin x)^2}$$

$$=2\sqrt{\cos x+\sin x} \quad \Rightarrow \quad \cos x+\sin x=0$$

$$\Rightarrow \quad \tan x=-1 \quad \Rightarrow \quad x=n\pi-\frac{\pi}{4}$$

Now,

$$\sqrt{\cos x-\sin x}\left\{\sqrt{\cos x-\sin x}+\sqrt{\cos x+\sin x}-2\right\}=0$$

$$\Rightarrow \quad \cos x+\sqrt{\cos2x}=2$$

$$\Rightarrow \quad \cos^2 x+4\cos x-5=0 \quad \Rightarrow \quad \cos x=1$$

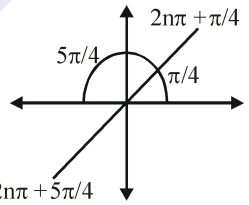
$$\Rightarrow \quad x=2n\pi$$

12. Ans. (A,C)

(i) when $\tan x>0$

$\tan^2 x=1 \Rightarrow \tan x=1$

$$x=n\pi+\frac{\pi}{4}$$



(ii) when $\tan x<0$

$\therefore -\tan x=\cot x$

$\Rightarrow \tan^2(x)=-1$ not possible

13. Ans. (A,C)

$$\frac{(\sqrt{3}-1)\cos\theta}{2\sqrt{2}}+\frac{(\sqrt{3}+1)\sin\theta}{2\sqrt{2}}=\frac{(\sqrt{3}-1)}{2\sqrt{2}}$$

$$\cos\left(\theta-\frac{5\pi}{12}\right)=\cos\frac{5\pi}{12} \quad \Rightarrow \quad \theta-\frac{5\pi}{12}=2n\pi\pm\frac{5\pi}{12}$$

$$\Rightarrow \quad 2n\pi+\frac{5\pi}{6} \quad \text{or} \quad 2n\pi$$

14. Ans. (A,D)

$2x=\cos(\alpha-\beta-\gamma+\delta)-\cos(\alpha-\beta+\gamma-\delta)$

$2y=\cos(\beta-\gamma-\alpha+\delta)-\cos(\beta-\gamma+\alpha-\delta)$

and similarly $2z=\cos(\gamma-\alpha-\beta+\delta)-\cos(\gamma-\alpha+\beta-\delta)$

Adding, $2x+2y+2z=0 \Rightarrow x+y+z=0$

$\therefore \quad x^3+y^3+z^3-3xyz=0$

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15. Ans. (B,D)

Sol. $\log_{x+1}(x-0.5) = \log_{x-0.5}(x+1)$

$\Rightarrow \left[\log_{x+1}\left(x-\frac{1}{2}\right)\right]^2 = 1 \Rightarrow \log_{x+1}\left(x-\frac{1}{2}\right) = 1$ or -1

$\Rightarrow x-\frac{1}{2} = (x+1)$ or $(x+1)^{-1} = x-\frac{1}{2}$

$\Rightarrow x = 1, x = -\frac{3}{2}$ (Reject)

16. Ans. (A,B,C)

Sol. $\log_{10}(1.96-1) < 0$ Hence (A) is incorrect

$a^2 - a - 2 = 0 \Rightarrow a = 2$ or $\underset{\text{rej}}{-1}$

Hence (B) is incorrect

$m = 7$ & $n = 7^4$ Hence (D) is correct

$\log_{0.1}(\sqrt{2}-1) > 0$ Hence (C) is incorrect

Hence (A) (B) (C)

SECTION–III

1. Ans. 7

$\because \angle A = 100^\circ \Rightarrow \angle B = \angle C = 40^\circ$

Now, $(x^2 + y^2) = [(\sin A + \sin B + \sin C)^2 + (\cos A + \cos B + \cos C)^2]$

$= 1 + 1 + 1 + 2\sin A\sin B + 2\sin B\sin C + 2\sin C\sin A + 2\cos A\cos B + 2\cos B\cos C + 2\cos C\cos A$

$= 3 + 2\{(\cos A\cos B + \sin A\sin B) + (\cos B\cos C + \sin B\sin C) + (\cos A\cos C + \sin A\sin C)\}$

$= 3 + 2\{\cos 60^\circ + \cos 0^\circ + \cos(-60^\circ)\}$

$= 3 + 2\left\{\frac{1}{2} + 1 + \frac{1}{2}\right\} = 7$

2. Ans. 0

$$\sum_{r=1}^{89}\log_{10}\tan\left(\frac{\pi r}{180}\right) = \left(\log_{10}\tan\frac{\pi}{180} + \log_{10}\tan\frac{89\pi}{180}\right)$$

$$+\left(\log_{10}\tan\frac{2\pi}{180} + \log_{10}\tan\frac{88\pi}{180}\right) + \ldots\ldots$$

$$= \left(\log_{10}\tan\frac{\pi}{180} - \log_{10}\tan\frac{\pi}{180}\right)$$

$$+\left(\log_{10}\tan\frac{2\pi}{180} - \log_{10}\tan\frac{2\pi}{180}\right) + \ldots\ldots$$

$= 0$

3. Ans. 7

Sol. $\ell n\left(\frac{a+b}{3}\right) = \left(\frac{\ell na + \ell nb}{2}\right)$

$\Rightarrow \ell n\left(\frac{a+b}{3}\right) = \ell n(ab)^{1/2}$

$\Rightarrow \left(\frac{a+b}{3}\right)^2 = (ab)$

$\Rightarrow a^2 + b^2 = 7ab$

$\Rightarrow \frac{a}{b} + \frac{b}{a} = 7$

4. Ans. 3

$\underbrace{\cos^4 x}_{\le 1} = \underbrace{1+\sin^{36} x}_{\ge 1}$ only equality holds

$\cos^2 x = 1$ and $\sin(x) = 0$

$x = n\pi$

solution are $0, \pi, 2\pi$

5. Ans. 4

$1 + |\sin x| + |\sin^2 x| + \ldots\ldots \infty = 2$

$\frac{1}{1-|\sin x|} = 2 \qquad 1-|\sin x| = \frac{1}{2}$

$|\sin x| = \frac{1}{2} \qquad \sin x = \pm\frac{1}{2}$

6. Ans. 6

$$\frac{1}{\sin\left(\frac{\pi}{4}+x\right)} + \frac{1}{\sin\left(\frac{\pi}{4}-x\right)} = 2\sqrt{2}$$

$$\frac{\sqrt{2}}{\cos x + \sin x} + \frac{\sqrt{2}}{(\cos x - \sin x)} = 2\sqrt{2}$$

$$\frac{\cos x}{\cos^2 x - \sin^2 x} = 1$$

$\cos x = 2\cos^2 x - 1$

$2\cos^2 x - \cos x - 1 = 0$

$2\cos^2 x - 2\cos x + \cos x - 1 = 0$

$\cos x = 1, \cos x = -\frac{1}{2}$

$\lambda_1 = 0, \ \lambda_2 = \frac{2\pi}{3}$

$\sin^2\lambda_1 + \sin^2\lambda_2 = 0 + \frac{3}{4}$

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